Hand Break

Three-minute hand and wrist breaks for keyboard workers, with progress you can actually see.

Productivity track | New Experience | Target: Best New Productivity Experience

AT A GLANCE

Track / division	Productivity (a habit tied to a recurring context: the work break) / New Experience
Primary prize	Best New Productivity Experience ($70,000; runner-up $35,000)
Fallback special award	Best Reason to Come Back ($25,000)
Session length	3 minutes, 5 movements
Room requirement	A desk surface, used as a reference plane for the forearm and fingers
Core tech	WebXR hand tracking joint angles, plane detection, IndexedDB trend storage
Biggest risk	Health claims, and a weak answer to 'why put on a headset for this?'
Solo feasibility	The easiest of the six ideas to build

The one-line pitch

A short, playful hand-mobility break for people who type all day. Hand tracking guides you through gentle finger and wrist movements at your desk, turns each one into a small interaction, and measures your range of motion so you can watch it change over weeks.

SECTION 01

The idea

People who type all day often end the day with stiff fingers and tired wrists. General advice is to take short movement breaks, but few people do them, and nobody can tell whether their mobility is improving.

Hand Break is a 3-minute routine of gentle hand and finger movements performed at your desk. Each movement is a tiny interaction: spreading your fingers opens a flower, touching each fingertip to your thumb plays a note of a melody, and a sequence of finger bends grows a vine. Because the headset tracks every joint, Hand Break also measures how far your fingers bend and spread, and shows the trend across weeks.

Important: wellness, not medicine

Hand Break must not claim to diagnose, treat or prevent any condition. It offers general mobility movements, tells users to stop if anything hurts, and recommends seeing a professional for pain. Ideally, a physiotherapist or occupational therapist reviews the movement list (credited in the submission). Measurements are shown as personal trends, never as clinical values.

SECTION 02

The routine

#	Movement	Interaction	What is measured
1	Finger spread and close (both hands)	A flower on the desk opens as you spread and closes as you make a relaxed fist.	Span from thumb tip to little fingertip; spread angle between fingers.
2	Tendon glide sequence: straight hand, hook, full fist, tabletop, straight fist	A vine climbs one step with each correct position.	Bend at each finger joint for each position.
3	Thumb touches: thumb to each fingertip in turn	Each touch plays the next note of a short melody.	Time to complete; accuracy of each touch.
4	Finger lifts with the hand flat on the desk	Lift one finger at a time while the others stay on the desk; each lift lights a key.	Height of each lift above the desk; how still the other fingers stay.
5	Wrist bends with the forearm resting on the desk	A wave rolls across the desk as you tilt the hand up and down.	Wrist bend angle relative to the desk (and so, the resting forearm).

The routine rotates daily among a set of about 12 movements so it stays fresh, always keeping 5 per session and always within a gentle range.

SECTION 03

The key design idea: the desk as a measuring tool

Hand tracking reports the hand's joints but not the forearm, which makes wrist angles hard to measure. Hand Break solves this by asking the user to rest the forearm flat on the desk. The detected desk plane then stands in for the forearm, so wrist bend can be measured as the angle between the back of the hand and the desk.

The desk also gives a clean reference for finger lifts: the height of each fingertip above the plane is easy to measure. This makes the real desk functionally necessary, which answers the judges' question about purposeful passthrough better than decoration would.

SECTION 04

Why passthrough and the real room matter

- **The desk is the measuring reference.** Wrist and finger-lift measurements need the real desk plane.
- **It happens where you work.** The routine appears on your real desk, next to your real keyboard, which ties it to the working day.
- **You see your real hands.** Seeing your own hands move, with light overlays, is more comfortable and more trustworthy than avatar hands for a body-focused activity.

> **Honest assessment**
>
> The biggest weakness is the reason to wear a headset. Most people would not put on a Quest just for a 3-minute stretch. The strongest framing is for people who already work in their headset (virtual monitors, passthrough work setups): Hand Break becomes the break inside a session they are already in. Even so, switching from a work app to a WebXR session in the browser is a context switch, and judges may notice.

SECTION 05

The first five minutes

Time	What happens
0:00	Passthrough. A small seed pot appears on the desk. Caption: 'Rest your forearms on the desk.'
0:20	A short, friendly safety note: gentle movements only, stop if anything hurts. One pinch to continue.
0:40	Movement 1: spread your fingers. The flower opens. The first measurement is recorded silently as your baseline.
1:30	Movements 2 to 5, each with a small reward.
3:30	Summary: a hand outline with today's values, and the seed pot now shows a sprout. 'Come back tomorrow and see it grow.'

SECTION 06

Progression and reasons to come back

System	Description
The desk plant	A plant on your desk grows with each completed break and blooms after a streak. It stays in the same spot using a spatial anchor.
Hand map	A visual of both hands where each finger's colour shows its range of motion trend over weeks.
Break goal	A daily goal (for example 3 breaks) with gentle, non-nagging tracking.
Weekly summary	'Your finger spread improved 6% this week' style messages, always framed as personal trends.

SECTION 07

Accessibility

- Adjustable range targets, so users with limited mobility get meaningful goals.
- One-hand routines.
- Seated by design, arms supported by the desk.
- Captions for all audio, and high-contrast overlays.
- No time pressure: each movement waits for the user.

SECTION 08

Technical architecture (IWSDK)

System	Responsibility
DeskReferenceSystem	Detects the desk plane and checks that the forearms are resting on it.
JointAngleSystem	Computes joint bend and spread angles from the 25 joints per hand each frame, without allocating memory.
RoutineSystem	Runs the 5-movement routine as a sequence of steps with targets.
InteractionFxSystem	Drives the flower, vine, melody, keys and wave feedback.
MeasurementSystem	Records the best stable value per movement (median over a short window to reduce tracking noise).
TrendSystem	Stores sessions in IndexedDB and computes smoothed weekly trends.

Hand-tracking angle measurements can vary by several degrees from frame to frame, so the app uses medians over short windows and shows smoothed weekly trends rather than single-session numbers.

SECTION 09

Risks and mitigations

Risk	Severity	Mitigation
Health claims or unsafe movements	High	Wellness framing; professional review of movements; stop-if-pain guidance; gentle ranges only.
Weak reason to wear a headset	High	Target people who already work in a headset; make the desk measurement essential.
Measurement noise	Medium	Median filtering; trends over weeks; never show clinical-looking values.
Low novelty for judges	Medium	Invest in delightful interactions (melody, vine, wave) and visible growth.
Reminders are hard from a web app	Low	Rely on the user's own routine; optional phone reminder link as a stretch goal.

SECTION 10

Seven-week build plan

Week	Dates	Goal	Exit criterion
Week 1	29 Sep - 4 Oct	Joint-angle measurement spike; desk plane reference; wrist angle test	Stable, repeatable measurements on device
Week 2	5 - 11 Oct	Routine engine; movements 1 and 2 with feedback	Two movements playable end to end
Week 3	12 - 18 Oct	Movements 3 to 5; safety flow; professional review requested	Full routine playable
Week 4	19 - 25 Oct	Trend storage; hand map; desk plant	Progress visible across sessions
Week 5	26 Oct - 1 Nov	Movement rotation (12 movements); accessibility options; sound	Content complete
Week 6	2 - 8 Nov	Performance; playtests with 5+ desk workers over several days	Testers return on their own
Week 7	9 - 15 Nov	Polish; record video; write submission	Video and text ready
Ship	16 - 18 Nov	Final checks; submit on 17 Nov	Submitted early

SECTION 11

Scope

Must ship	Nice to have	Cut
5-movement routine; desk reference; measurements; trends; desk plant; safety flow	12-movement rotation; hand map visual; weekly summary	Reminders; integrations with work apps; any diagnostic features

SECTION 12

Demo video plan (under 3 minutes)

Time	Beat	Shot
0:00 - 0:15	Hook	Fingers touch a thumb one by one and a melody plays over a real desk and keyboard.
0:15 - 0:45	The problem	Long typing days; nobody takes breaks; nobody knows if they are improving.
0:45 - 1:50	A full break	The 3-minute routine in real time, condensed.
1:50 - 2:30	Progress	The hand map and desk plant after three weeks.
2:30 - 2:55	Close	Back to work in the same headset session.

SECTION 13

Submission draft

Name	Hand Break
Tagline (under 140 characters)	Three-minute hand and wrist breaks for keyboard workers, with progress you can actually see.
Track / division	Productivity / New Experience
Target launch	Web release Q1 2027

Description (first draft)

Typing all day is hard on hands, and short movement breaks are easy to skip. Hand Break makes a 3-minute hand break something you want to do, and shows you it is working.

At your real desk, gentle finger and wrist movements become small interactions: spreading your fingers opens a flower, touching each fingertip to your thumb plays a melody, and resting your forearm on the desk lets you roll a wave with your wrist. Hand tracking measures how far each finger moves, and a plant on your desk and a colour map of your hands show the trend over weeks.

We built Hand Break with the Immersive Web SDK, using the desk plane as a reference for wrist and finger measurements. It is a wellness tool, not a medical device.

SECTION 14

How it maps to the judging criteria

Criterion (25% each)	Evidence in Hand Break
Innovation and Creativity	Hand tracking used as a measurement instrument; the desk as the forearm reference.
Experience Design	A recurring daily moment, 3-minute sessions, delightful feedback. The 'why a headset' question is the weak spot.
Technical Implementation	Robust joint-angle measurement with noise handling.
Polish and Presentation	Calm, clean visuals; clear value.

Hard requirement and design guidelines

Test	What the rules ask	How this idea passes
Hands-first (hard requirement)	Complete the entire experience without ever pairing a controller.	The hands are both the input and the subject; no controllers anywhere.
Airplane seat test	Every interaction works within a two-foot radius, seated.	All movements are done seated with forearms on the desk.
One bus stop test	Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less.	One break is 3 minutes and complete.
Take it away test	Remove any third-party service and a project still stands.	No third-party services.

Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.

- **Policy:** avoid any health claims in the app, video and description; follow Meta's developer content policies on health and wellness.

SECTION 15

Honest weaknesses

- The 'why wear a headset for this?' question is hard to answer convincingly.
- Novelty is low compared with the other ideas.
- Health-adjacent products carry claim and safety risks.
- The Productivity prize is lower than Gaming and Entertainment.

SECTION 16

Scorecard and verdict

Dimension	Score / 10	Comment
Originality	5	Hand exercise apps exist; measurement via the desk is the new part.
Only possible in MR	4	The desk reference helps, but much of it would work elsewhere.
Works in untested rooms	10	Needs only a desk.
Solo feasibility in 7 weeks	9	The simplest build.
Prize ceiling	$70k	Productivity track.

Verdict

The safest build but the weakest competitive position. A good choice only if you need a low-risk project; otherwise its best parts (joint measurement, the desk as reference) could become a feature inside another idea.